

PRENTISS, AB, Canada

WMO: 713530

Lat: 52.4336N

Lon: 113.5921W

Elev: 965

StdP: 90.26

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.7	-24.8	-31.4	0.2	-27.6	-28.3	0.3	-24.6	11.3	-2.8	9.8	-4.6	3.3	280	0.679

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	27.1	17.0	25.3	15.9	23.7	15.2	18.7	24.9	17.4	23.5	16.3	22.0	3.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.3	13.1	22.5	14.9	11.9	20.7	13.7	11.0	19.5	57.2	25.0	53.0	23.6	49.2	22.1	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.0	7.9	-30.8	30.5	2.8	1.9	-32.8	31.9	-34.5	33.0	-36.0	34.1	-38.1	35.5		
(5)			-31.0	21.1	2.8	1.2	-33.0	21.9	-34.6	22.6	-36.2	23.3	-38.2	24.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	-7.9	-8.9	-2.7	3.8	10.3	13.9	16.4	15.2	10.9	4.6	-3.2	-7.8	(6)
(7)		DBStd	10.89	8.46	8.12	6.95	5.32	4.52	3.02	2.93	3.31	4.70	5.24	7.14	7.54	(7)
(8)		HDD10.0	2898	554	529	393	193	51	4	0	3	43	179	396	553	(8)
(9)		HDD18.3	5336	812	762	651	436	250	137	73	106	224	427	645	812	(9)
(10)		CDD10.0	636	0	0	1	7	61	120	200	163	71	11	1	0	(10)
(11)		CDD18.3	31	0	0	0	0	2	3	15	8	3	0	0	0	(11)
(12)		CDH23.3	429	0	0	0	1	37	39	164	126	58	3	0	0	(12)
(13)		CDH26.7	66	0	0	0	0	3	3	28	21	11	0	0	0	(13)
(14)	Wind	WSAvg	3.9	4.2	3.9	4.0	4.4	4.4	3.9	3.3	3.2	3.6	4.0	4.0	4.1	(14)
(15)	Precipitation	PrecAvg	485	22	15	21	28	58	94	87	64	36	22	20	18	(15)
(16)		PrecMax	591	64	40	50	74	119	178	191	117	79	63	60	43	(16)
(17)		PrecMin	360	4	4	4	5	23	21	17	19	4	7	3	4	(17)
(18)		PrecStd	61	13	8	10	16	22	33	37	24	21	13	13	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.1	8.8	13.5	21.7	26.7	26.2	29.4	28.8	28.3	22.5	15.1	8.3	(19)
(20)			MCWB	2.9	3.5	6.6	9.8	12.4	16.2	19.3	16.8	14.6	11.1	7.7	3.3	(20)
(21)		2%	DB	6.4	5.6	9.8	18.6	24.0	24.5	27.2	26.4	24.8	17.9	10.7	6.0	(21)
(22)			MCWB	2.3	1.2	4.2	8.7	11.7	15.1	18.9	16.8	13.9	9.7	5.1	1.9	(22)
(23)		5%	DB	5.0	3.7	7.5	15.1	21.6	22.5	25.0	24.5	22.0	14.9	7.4	4.2	(23)
(24)			MCWB	1.3	0.1	3.0	6.5	10.8	14.1	17.6	16.3	13.0	8.0	2.8	0.6	(24)
(25)		10%	DB	3.0	1.8	5.6	12.2	19.0	20.6	23.2	22.4	19.1	12.3	5.1	2.1	(25)
(26)			MCWB	-0.1	-1.1	1.8	5.1	9.9	13.1	16.4	15.6	11.8	6.5	1.4	-1.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.8	6.8	10.4	13.8	18.2	21.3	19.2	16.2	12.0	8.0	3.7	(27)
(28)			MCDWB	7.1	8.5	13.0	21.0	23.5	24.8	26.8	25.6	25.0	21.1	14.5	7.6	(28)
(29)		2%	WB	2.5	1.6	4.6	8.8	12.3	16.3	19.5	18.1	14.5	9.9	5.3	2.1	(29)
(30)			MCDWB	6.0	5.1	9.0	17.9	22.0	22.3	25.6	24.5	22.8	17.5	10.0	5.8	(30)
(31)		5%	WB	1.4	0.3	3.1	7.1	11.3	15.0	18.1	16.9	13.3	8.4	3.3	0.7	(31)
(32)			MCDWB	4.8	3.3	7.2	14.1	19.9	20.8	24.1	23.1	20.8	14.2	6.7	4.0	(32)
(33)		10%	WB	0.0	-1.0	2.0	5.6	10.3	13.8	16.9	15.7	12.1	6.9	1.8	-0.9	(33)
(34)			MCDWB	2.9	1.8	5.4	11.4	17.8	19.3	22.4	21.8	18.5	11.7	4.6	1.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.5	8.9	8.8	10.5	11.7	10.4	11.2	11.5	11.4	10.1	8.2	7.7	(35)
(36)			MCDBR	8.9	10.2	11.2	15.6	15.8	13.4	13.8	15.2	16.2	14.3	11.0	8.6	(36)
(37)			MCWBR	6.2	7.0	7.2	8.1	6.9	6.4	7.6	7.8	7.6	7.7	6.9	6.1	(37)
(38)		5% WB	MCDBR	8.7	10.1	10.8	14.6	14.2	11.9	13.3	14.0	14.8	14.0	10.4	8.5	(38)
(39)			MCWBR	6.3	7.2	7.2	7.8	6.4	6.5	7.7	7.8	7.3	7.7	6.9	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.257	0.262	0.288	0.318	0.331	0.338	0.340	0.352	0.303	0.278	0.250	0.240	(40)	
(41)		taud	2.321	2.397	2.425	2.391	2.410	2.413	2.434	2.374	2.491	2.544	2.477	2.328	(41)	
(42)		Ebn at Noon	756	874	915	920	920	913	905	873	887	850	780	715	(42)	
(43)		Edh at Noon	59	74	89	105	109	110	106	106	84	64	50	49	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.07	2.11	3.51	4.71	5.72	6.05	6.35	5.19	3.61	2.16	1.10	0.79	(44)	
(45)		RadStd	0.10	0.17	0.26	0.43	0.36	0.39	0.34	0.35	0.39	0.25	0.11	0.07	(45)	

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(m)
(47) Regional (79 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page